

# MoLAG: monocular LED assignment via a graph neural network for multi-tracker surgical navigation

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## Abstract

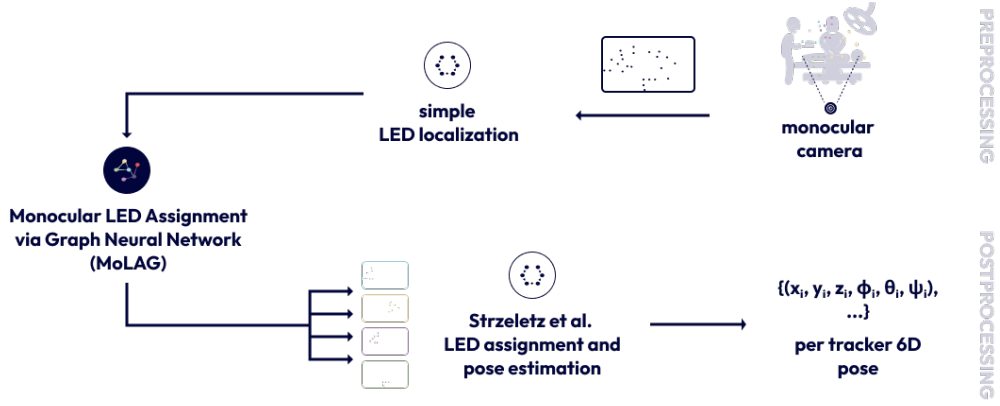
**Purpose:** Monocular peer-to-peer tracking must group an unordered, scene-wide set of localised light-emitting diode (LED) points by source tracker before tracker-specific LED identification and point-to-line pose estimation. MoLAG applies a graph neural network to this grouping task.

**Methods:** MoLAG uses complete directed message passing and a symmetric affinity head to assign a same-tracker score to every unordered pair of detected points within a scene. Thresholded affinities form candidate tracker components. A scaled-conjunction loss promotes within-tracker connectivity, between-tracker separation, and suppression of spurious connections. Six full-fidelity loss settings were trained on generated coordinate scenes and evaluated on a frozen 50,000-scene synthetic test suite. A separate replicated hyperparameter search explored the loss family at reduced training fidelity.

**Results:** The combined full-fidelity setting achieved 89.64 % complete-tracker extractability [89.35, 89.94] among 51,817 complete seven-LED trackers. Strict scene grouping was 56.43 % [55.99, 56.86], all-tracker strict recovery was 76.20 %, and real-real edge  $F_1$  was 92.34 %. Among the six full-fidelity settings, the combined setting produced the highest measured strict grouping and extractability. The reduced-fidelity search identified a replicated high-recall operating region.

**Conclusion:** MoLAG effectively partitions multi-tracker LED detections on the evaluated synthetic coordinate distribution and retains most complete constellations for subsequent geometry-specific processing.

**Keywords:** surgical navigation, optical tracking, LED tracker grouping, graph neural network, affinity learning, synthetic data



**Fig. 1** MoLAG in the tracking pipeline. Preprocessing localises 2D blob centres. MoLAG partitions this scene-wide set into candidate tracker groups. The method of Strzeletz et al. [15] then identifies the LEDs and estimates each pose by point-to-line optimisation

## 1 Introduction

Optical tracking is widely used in surgical navigation [3, 11]. Commercial systems observe rigid marker arrays attached to instruments and patient reference frames. Stereo systems triangulate the markers and register them to the known geometry, making pose availability dependent on marker visibility and array design [6, 20].

Strzeletz et al. [15] proposed a peer-to-peer architecture in which each device acts as both a tracked object and a tracking device, using a monocular camera to observe nearby devices. Each frame yields an unordered set of 2D detections from all visible trackers, possibly including spurious points. Tracker membership and LED identity are unknown. Their rule-based method forms marker-triplet hypotheses over the complete set, whose number grows cubically with the number of detections, and requires a complete seven-LED constellation for LED identification and pose estimation. Crowding therefore enlarges the search, while missing detections reduce tracker availability.

*MoLAG* (**M**onocular **L**ED **A**ssignment via **G**raph neural network) partitions these detections by source tracker before tracker-specific LED identification and pose estimation (Figure 1). Complete directed message passing provides scene context, a symmetric head predicts same-tracker affinities for every unordered detection pair, and thresholded connected components form the groups. Training uses generated coordinate sets with perspective variation, missing LEDs, spurious detections, and localisation noise rather than camera images.

The contributions are (1) a geometry-independent affinity architecture for variable-size multi-tracker point sets; (2) a scaled-conjunction loss that promotes within-tracker connectivity, tracker separation, and control of spurious connections; and (3) a full-fidelity loss study, replicated proxy-fidelity hyperparameter search, and detailed analysis of grouping performance on the evaluated synthetic distribution.

## 2 Related work

Monocular optical tracking reduces the camera-system footprint. Recent approaches use retroreflective fiducials [12], estimate instrument pose from images [9], or decode temporally modulated LEDs with an event camera [4]. These methods localise points, infer pose directly, or resolve LED identity through hardware signals. MoLAG instead groups an existing, unlabelled set of point detections.

The intended downstream method identifies geometric marker triplets and estimates pose by Lagrange-Newton point-to-line optimisation [14, 15]. Its square seven-LED marker differs from the triangular geometry evaluated here. Geometric LED correspondence also appears in monocular aerial tracking [5, 16]. These methods encode a specific marker layout, whereas MoLAG predicts tracker membership before geometry-specific processing.

MoLAG combines two established ideas. DGCNN introduced EdgeConv for message passing on point sets [19], and neural association methods predict pairwise relations before forming groups or instances [2, 18]. MoLAG uses a complete directed graph for contextual node embeddings and a symmetric affinity head for unordered detection pairs.

Correlation clustering and multicut methods enforce a globally consistent partition of pair relations [1, 13]. MoLAG instead uses direct thresholding and connected components for lightweight group extraction.

Independent affinity errors need not preserve the partition obtained after thresholding. Maximin affinity learning therefore trains graph connectivity rather than isolated edges [8, 17], while spanning-tree objectives retain within-class connectivity with few required edges [21]. The scaled-conjunction objective adopts these principles for tracker connectivity and separation, with additional terms for spurious attachments and paths.

## 3 Method

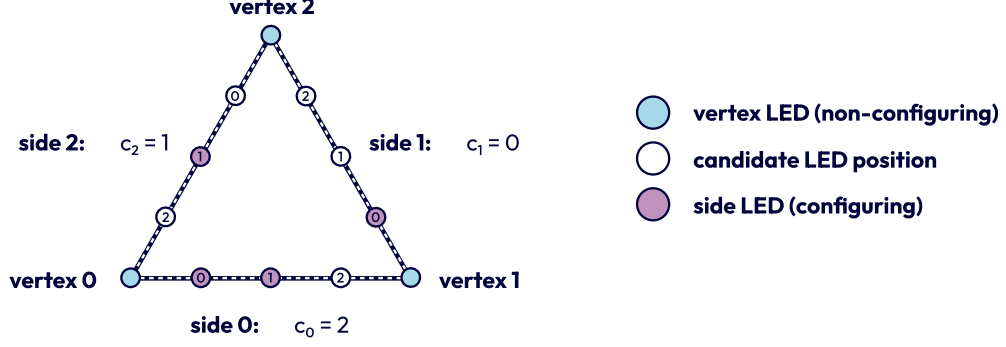
### 3.1 Task and evaluated tracker

For one monocular frame, MoLAG receives an unordered set of localised points

$$\mathcal{X} = \{\mathbf{x}_i\}_{i=1}^N, \quad \mathbf{x}_i \in \mathbb{R}^2, \quad (1)$$

containing LEDs from several trackers and possible spurious detections. It predicts a same-tracker affinity for every unordered pair of detected points and returns connected components as candidate groups. A downstream method can then identify LEDs and estimate pose within each group. The grouping formulation depends only on coordinates and tracker-membership labels, not a marker template.

The experiments use a triangular 64-mm tracker with seven LEDs (Figure 2). Three LEDs occupy the vertices. A code  $\mathbf{c} = (c_0, c_1, c_2)$  selects four side LEDs from three candidate positions per side:  $c_1$  and  $c_2$  select one position each, while  $c_0$  identifies the omitted position among two occupied locations. This asymmetric encoding gives



**Fig. 2** Evaluated seven-LED tracker for code  $\mathbf{c} = (2, 0, 1)$ . Three fixed vertex LEDs and four code-dependent side LEDs form one of 27 configurations

$3^3 = 27$  configurations. Other layouts require representative training data but no architectural change.

### 3.2 Coordinate-level scene generation

Training scenes are generated at MoLAG’s image-point interface. Rigid 3D trackers are sampled, projected through a monocular pinhole camera, and modified by LED dropout, spurious points, and Gaussian coordinate noise. Inter-tracker body occlusion is rejected geometrically. Each point set is mean-centred and divided by its maximum radius,  $\tilde{\mathbf{x}}_i = (\mathbf{x}_i - \bar{\mathbf{x}}) / \max_j \|\mathbf{x}_j - \bar{\mathbf{x}}\|_2$ . This removes translation and uniform scale while preserving relative geometry. The exact camera, pose, and error distributions are reported in Supplementary Section S2.2.

### 3.3 Affinity graph network

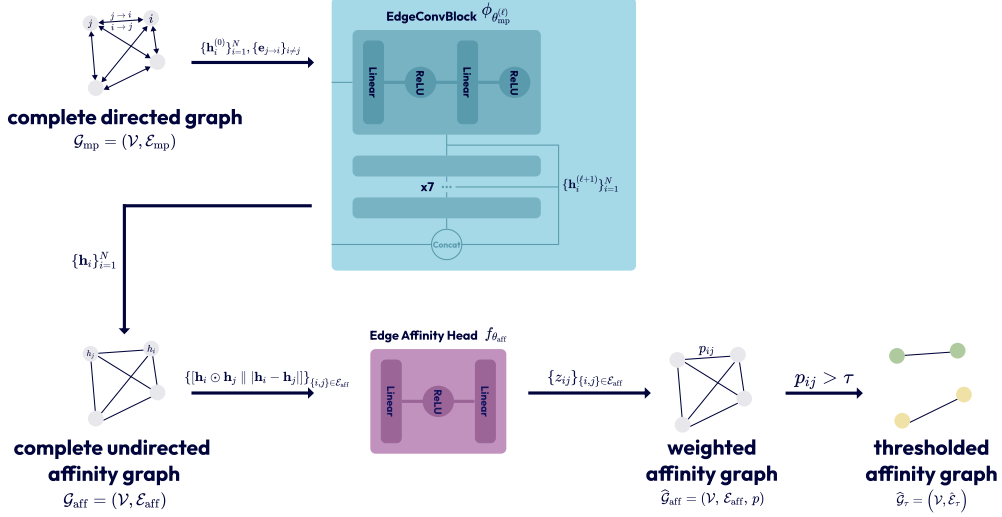
MoLAG uses a complete directed message-passing graph: every point receives information from every other point in the frame. Seven EdgeConv-inspired blocks [19] combine endpoint features with squared distance and direction. The seven block outputs are concatenated into a node embedding  $\mathbf{h}_i$ . For each unordered pair  $\{i, j\}$  of detected points, a two-layer multilayer perceptron evaluates the symmetric representation

$$z_{ij} = f_{\theta}([\mathbf{h}_i \odot \mathbf{h}_j \parallel \|\mathbf{h}_i - \mathbf{h}_j\|]), \quad p_{ij} = \sigma(z_{ij}). \quad (2)$$

The Hadamard product and absolute difference are invariant to exchanging the two points. Figure 3 summarises the architecture; layer equations and dimensions appear in Supplementary Section S2.1.

### 3.4 Scaled-conjunction affinity objective

For scene  $s$ ,  $V_{s,t}$  contains the visible nodes of tracker  $t$ ,  $\mathcal{T}_s$  the tracker set, and  $B_s$  the spurious nodes. Independent pair supervision does not directly encode the required partition: each tracker needs only enough positive edges to remain connected, while



**Fig. 3** MoLAG architecture. Complete directed message passing forms node embeddings. A symmetric head predicts affinities for unordered pairs, and connected components of the thresholded graph define candidate groups

one cross-tracker edge can merge two components. The structured objective therefore combines tracker connectivity, tracker-pair separation, spurious attachment, and spurious-bridge conditions.

Within a condition, edge losses  $v_a$  with non-negative handicaps  $\eta_a$  are reduced by the weighted soft maximum

$$\mathcal{M}_\beta(v, \eta; A) = \frac{1}{\beta} \left[ \log \sum_{a \in A} e^{\beta(v_a - \eta_a)} - \log \sum_{a \in A} e^{-\beta \eta_a} \right]. \quad (3)$$

Finite  $\beta$  distributes gradient across poor edges;  $\beta \rightarrow \infty$  recovers the worst handicapped edge. If scene  $s$  contains  $n_s$  applicable conditions with losses  $q_{s,\kappa}$ , the scene reduction is

$$\mathcal{R}_{p,\rho}(q) = \frac{\sum_{s \in \mathcal{S}^*} n_s^p \left( n_s^{-1} \sum_{\kappa=1}^{n_s} q_{s,\kappa} \right)}{(1-\rho)S + \rho|\mathcal{S}^*|}, \quad \mathcal{S}^* = \{s : n_s > 0\}. \quad (4)$$

The power  $p \in [0, 1]$  controls the influence of crowded scenes, and  $\rho$  determines whether scenes without an applicable condition enter the outer mean.

The connectivity term uses a maximum-affinity spanning tree for each tracker. The separation term penalises the strongest edge between each tracker pair. The attachment term penalises the strongest link from a spurious point to a tracker, and the bridge term penalises the widest path between trackers through spurious nodes. An optional scene-local supervised contrastive term structures real-node embeddings [10]. The complete loss is

$$\mathcal{L} = w_c \mathcal{L}_{\text{conn}} + w_s (\mathcal{L}_{\text{pair}} + \varepsilon_{\text{spur}} \mathcal{L}_{\text{att}}) + w_b \mathcal{L}_{\text{bridge}} + w_{sc} \mathcal{L}_{\text{supcon}}. \quad (5)$$

Supplementary Section S3 defines every term and boundary case. The combined full-fidelity setting uses  $w_c = 1$ ,  $w_s = 0.46$ ,  $w_b = 0.25$ ,  $m_c = m_s = 1$ ,  $m_b = 0$ ,  $\beta = 1$ ,  $\delta = 3$ ,  $\varepsilon_{\text{spur}} = 0.01$ ,  $p_c = p_s = 0.5$ ,  $\rho = 1$ , and  $w_{sc} = 0$ .

### 3.5 Group extraction

At inference, affinities  $p_{ij} \geq \tau$  are retained, where  $\tau$  is selected on held-out calibration data. Connected components form the candidate groups; isolated points form singleton components. The resulting partition is deterministic given the network predictions.

## 4 Experiments

### 4.1 Training and evaluation protocol

The study evaluates overall grouping, six full-fidelity loss configurations, a proxy-fidelity hyperparameter search, and performance across scene compositions. Each full-fidelity run used the same architecture and 5,000,000 generated scenes for ten epochs. Scenes contained one to ten distinct tracker configurations. The generator sampled tracker centres within  $x, y \in [-200, 200]$  mm and  $z \in [100, 600]$  mm, face tilts up to  $85^\circ$ , LED dropout, up to four spurious detections, and coordinate noise. Supplementary Sections S2.2 and S2.1 report all generation and training parameters.

Each checkpoint was calibrated on 5,000 frozen scenes, with 500 for every tracker count from one to ten. Thresholds from 0.05 to 0.95 were evaluated in steps of 0.01, and the threshold with the highest strict grouping accuracy was retained. Final evaluation used a disjoint frozen suite of 50,000 scenes, stratified into 5,000 scenes for each exact tracker count. Training, in-training evaluation, calibration, HPO validation, and final test scenes were mutually disjoint. All checkpoints used the same test scenes, enabling paired comparisons.

### 4.2 Evaluation metrics

Let  $\hat{c}_s(i)$  denote the predicted component of node  $i$  in scene  $s$ . Tracker  $t$  is strictly recovered when all nodes in  $V_{s,t}$  share one component and that component contains no real node from another tracker. Let  $r_{s,t}$  indicate this event. For  $N$  test scenes, strict scene grouping and all-tracker strict recovery are

$$A_{\text{group}} = \frac{1}{N} \sum_s \prod_{t \in \mathcal{T}_s} r_{s,t}, \quad R_{\text{all}}^{\text{strict}} = \frac{\sum_s \sum_{t \in \mathcal{T}_s} r_{s,t}}{\sum_s |\mathcal{T}_s|}. \quad (6)$$

A scene therefore succeeds only if every tracker is unsplit and unmerged. Spurious detections may attach to one tracker but cause failure if they connect trackers.

Complete trackers contain all seven LEDs. Complete-tracker strict recovery applies the same criterion to this subset. Complete-tracker extractability is the number of distinct predicted components containing at least one unsplit complete tracker, divided by

**Table 1** Changed loss parameters in the full-fidelity study. Unlisted parameters are fixed as stated in the text

| Configuration               | Mode        | $\beta$  | $\delta$ | $p_c$ | $p_s$ | Eligible | $w_s$ | $w_{sc}$ | $\epsilon_{\text{spur}}$ |
|-----------------------------|-------------|----------|----------|-------|-------|----------|-------|----------|--------------------------|
| Conjunction reference       | Conjunction | $\infty$ | $\infty$ | 0     | 0     | No       | 2.00  | 0.03     | 0                        |
| Dense gradient              | Scaled      | 1        | 3        | 0     | 0     | No       | 2.00  | 0.03     | 0                        |
| Reduced separation          | Conjunction | $\infty$ | $\infty$ | 0     | 0     | No       | 0.90  | 0.03     | 0                        |
| Scene weighting             | Scaled      | $\infty$ | $\infty$ | 0.5   | 0.5   | Yes      | 1.00  | 0.03     | 0                        |
| No SupCon                   | Conjunction | $\infty$ | $\infty$ | 0     | 0     | No       | 2.00  | 0        | 0                        |
| Combined scaled conjunction | Scaled      | 1        | 3        | 0.5   | 0.5   | Yes      | 0.46  | 0        | 0.01                     |

the number of complete trackers. For an unsplit tracker, let  $\hat{c}_{s,t}$  denote its component:

$$E_{\text{complete}} = \frac{\sum_s \left| \left\{ \hat{c}_{s,t} : t \in \mathcal{T}_s^{(7)}, |\{\hat{c}_s(i) : i \in V_{s,t}\}| = 1 \right\} \right|}{\sum_s |\mathcal{T}_s^{(7)}|}. \quad (7)$$

A split tracker contributes zero; a component containing two complete trackers contributes once. Extractability is thus a merge-tolerant grouping proxy, conditional on a downstream method being able to identify an intact constellation among additional detections.

Conventional precision, recall, and  $F_1$  quantify pairwise affinities. Real-only grouping removes edges incident to spurious nodes before component extraction. Further diagnostics distinguish splits, direct merges, mixed errors, spurious-mediated bridges, and spurious attachments. Supplementary Section S4 defines these diagnostics and gives illustrative cases.

### 4.3 Loss-configuration studies

The full-fidelity study trained one conjunction reference and five variants. All runs shared the network, data, optimiser, checkpoint rule, calibration, and test suite; Table 1 lists every changed loss parameter. Each of the first four variants addresses one design question relative to the reference. The combined scaled-conjunction configuration changes several parameters and is interpreted only as a complete setting. Across all rows,  $w_c = 1$ ,  $m_c = m_s = 1$ ,  $w_b = 0.25$ ,  $m_b = 0$ ,  $\tau_{sc} = 0.2$ , and the maximum tracker size is seven LEDs.

The exploratory search examined 77 settings at 6% training fidelity. Three retained candidates and the combined setting as proxy incumbent were then trained three times each. Threshold selection and scoring used disjoint halves of a frozen 20,000-scene suite, subject to a strict-grouping floor of 48.39%. This study measures replicated proxy-fidelity behaviour, not transfer to full-fidelity training. Its complete design and record appear in Supplementary Section S7.

**Table 2** Combined-configuration performance on the frozen test suite. Brackets contain 95 % intervals

| Measure                                   | Result                      |
|-------------------------------------------|-----------------------------|
| Strict scene grouping                     | 56.43 % [55.99, 56.86]      |
| Real-only grouping                        | 56.33 %                     |
| All-tracker strict recovery               | 76.20 %                     |
| Complete-tracker strict recovery          | 80.36 % [79.89, 80.83]      |
| Complete-tracker extractability           | 89.64 % [89.35, 89.94]      |
| All-pair edge $F_1$                       | 90.84 %                     |
| Real-real edge precision / recall / $F_1$ | 94.33 % / 90.43 % / 92.34 % |

#### 4.4 Statistical analysis

Strict scene grouping uses 95 % Wilson intervals. Complete-tracker ratios use scene-clustered delta-method intervals because tracker outcomes within a scene are correlated. Paired McNemar tests compare full-fidelity checkpoints on common binary scene outcomes; these single-run comparisons do not estimate training variation. The proxy study reports means and sample standard deviations across three runs per arm and uses pooled within-arm variation for its primary contrast. Supplementary Material reports the complete procedures and statistics.

## 5 Results

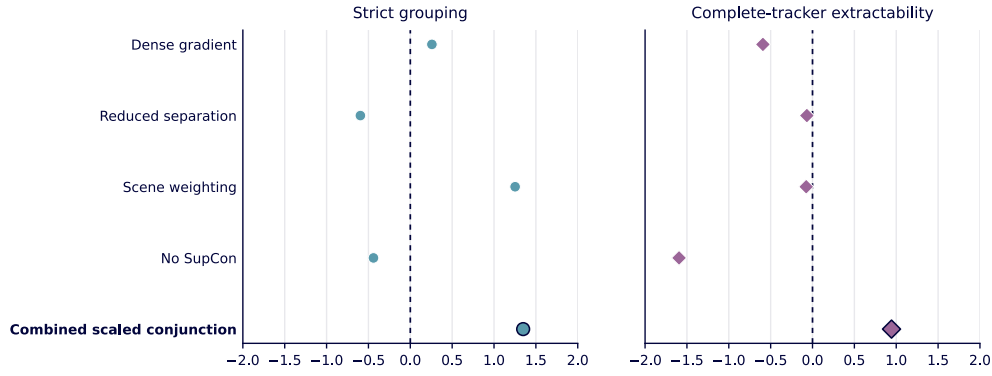
### 5.1 Overall performance

The test suite contained 51,817 complete trackers among 275,000 tracker instances. The combined full-fidelity configuration retained 89.64 % as extractable constellations. Complete-tracker strict recovery was 80.36 %, all-tracker strict recovery was 76.20 %, and 56.43 % of the 50,000 scenes were grouped exactly (Table 2). The difference between extractability and strict recovery shows that intact constellations often remained available inside imperfect partitions.

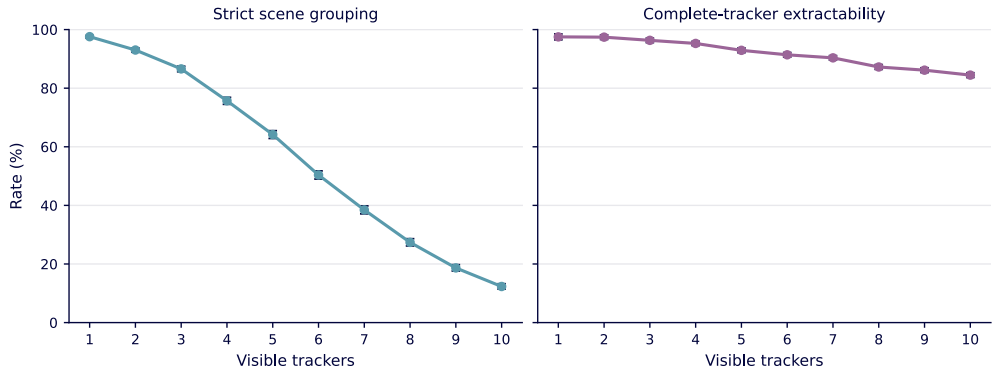
### 5.2 Loss configurations

The combined checkpoint led both primary outcomes among the six full-fidelity settings (Figure 4). Among the first four variants, scene weighting showed the largest increase in strict grouping while preserving extractability. Dense gradient sharing and reduced separation shifted the balance between the outcomes, whereas removing Sup-Con from the reference reduced both. Each setting was trained once, so these results compare checkpoints rather than training distributions; the combined setting does not isolate individual parameter effects. Supplementary Tables S1 and S2 report absolute results and paired statistics.

The combined setting subsequently served as the incumbent in the proxy-fidelity search. All three retained candidates exceeded its three-run mean extractability. In the



**Fig. 4** Percentage-point differences from the conjunction reference in strict scene grouping and complete-tracker extractability. Each point is one checkpoint



**Fig. 5** Strict scene grouping and complete-tracker extractability by exact visible tracker count. Each stratum contains 5,000 scenes; whiskers are 95 % intervals

primary comparison, which also retained an earlier incumbent run, the selected candidate improved extractability by 2.14 percentage points [1.21, 3.07] after accounting for measured between-run variation at this fidelity. Strict grouping and all-tracker strict recovery were similar, while edge  $F_1$  decreased and spurious attachment increased. No search-selected configuration was trained at full fidelity. Supplementary Section S7 reports the complete search and replication.

### 5.3 Scene composition and failures

Crowding affected exact grouping more strongly than extractability (Figure 5). Strict grouping fell from 97.58 % with one tracker to 12.32 % with ten, whereas complete-tracker extractability remained 84.46 % in the ten-tracker stratum. Merges and mixed split-and-merge errors dominated failed scenes; isolated splits and failures caused only by spurious bridges were rare. Visibility, spurious detections, tracker configuration, total point count, and the full failure composition are analysed in Supplementary Section S5.2.

## 6 Discussion

MoLAG partitions scene-wide detections before tracker-specific LED identification and pose estimation. Its 89.64 % extractability among 51,817 complete trackers is the central result for this role: most complete constellations remained intact in one candidate component, including in crowded scenes. Strict grouping was lower because it requires every tracker in a scene to be both unsplit and unmerged. The widening gap between these measures with tracker count identifies separation, rather than loss of complete constellations, as the main residual challenge.

The full-fidelity comparison supports the combined configuration as the strongest of the six evaluated checkpoints, but does not isolate the effects of its changed parameters. Independent repeats would be required for configuration-level claims. The replicated proxy-fidelity search identifies a stable high-extractability region, although lower edge precision and greater spurious attachment show the associated operating-point trade-off. These settings have not been tested at full fidelity.

The evidence applies to generated coordinate sets, not recorded images. The generator varies perspective, tracker count, dropout, clutter, and localisation noise while preserving exact membership labels. Its random dropout, uniform spurious points, Gaussian noise, ideal pinhole camera, and distinct configuration codes do not reproduce the spatial and appearance structure of an operating room. Translation and scale normalisation also remove global image position and apparent size. Transfer to real localisation output, repeated codes, and other tracker geometries therefore remains to be established.

The study evaluates independent frames and grouping outcomes. It does not measure temporal association, cross-camera fusion, runtime, LED identification, or pose accuracy. Extractability further assumes that the downstream geometric method can identify a complete constellation within a component containing additional detections; an equivalent procedure for the triangular tracker remains to be integrated. The number of predicted affinities grows quadratically with detected points, so the net computational effect depends on the downstream implementation. The single-run full-fidelity comparison and reused HPO selection split also leave training variation and selection optimism unresolved.

An end-to-end study should connect recorded LED localisation, MoLAG, tracker-specific identification, and point-to-line pose estimation. Measuring pose availability and accuracy, runtime, and localisation-error distributions will test the intended search reduction and support calibration of the synthetic training distribution.

## 7 Conclusion

MoLAG combines complete message passing, symmetric affinity prediction, and a scaled-conjunction loss to partition multi-tracker LED detections. On 50,000 generated test scenes, it retained 89.64 % of 51,817 complete trackers as extractable constellations, with 76.20 % all-tracker strict recovery and 56.43 % exact scene grouping. These results establish effective grouping on the evaluated coordinate distribution. The next step is end-to-end evaluation with recorded localisation output and downstream LED identification and pose estimation.

## Statements and Declarations

- **Funding:** Not applicable.
- **Competing interests:** The authors declare no competing interests.
- **Ethics approval:** Not applicable (synthetic data only).
- **Consent for publication:** Not applicable.
- **Data availability:** No patient or recorded camera data were used. All experimental input data were generated procedurally. The generation methodology and experiment configurations are available at <https://github.com/yannikheizmann/MoLAG>. Trained model weights and supporting benchmark and HPO artefacts are available on reasonable request.
- **Code availability:** The MoLAG implementation, including synthetic-data generation, training, threshold calibration, evaluation, and inference, is available with documentation and experiment configurations at <https://github.com/yannikheizmann/MoLAG>.
- **Author contributions:** Yannik Heizmann conceived the study, developed the method and software, designed and conducted the experiments, analysed and visualised the results, and wrote and revised the manuscript. Harald Hoppe contributed the initial project concept and reviewed the manuscript.

## References

- [1] Bansal N, Blum A, Chawla S (2004) Correlation clustering. *Machine Learning* 56(1–3):89–113. <https://doi.org/10.1023/B:MACH.0000033116.57574.95>
- [2] Brasó G, Leal-Taixé L (2020) Learning a neural solver for multiple object tracking. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp 6246–6256, <https://doi.org/10.1109/CVPR42600.2020.00628>
- [3] Cleary K, Peters TM (2010) Image-guided interventions: Technology review and clinical applications. *Annual Review of Biomedical Engineering* 12:119–142. <https://doi.org/10.1146/annurev-bioeng-070909-105249>
- [4] Ebmer G, Loch A, Vu MN, et al (2024) Real-time 6-DoF pose estimation by an event-based camera using active LED markers. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp 8122–8131, <https://doi.org/10.1109/WACV57701.2024.00795>
- [5] Faessler M, Mueggler E, Schwabe K, et al (2014) A monocular pose estimation system based on infrared LEDs. In: *2014 IEEE International Conference on Robotics and Automation (ICRA)*, pp 907–913, <https://doi.org/10.1109/ICRA.2014.6906962>
- [6] Fattori G, Lomax AJ, Weber DC, et al (2021) Technical assessment of the NDI Polaris Vega optical tracking system. *Radiation Oncology* 16(1):87. <https://doi.org/10.1186/s13014-021-01804-7>

- [7] Fey M, Lenssen JE (2019) Fast graph representation learning with PyTorch Geometric. In: ICLR 2019 Workshop on Representation Learning on Graphs and Manifolds, URL <https://arxiv.org/abs/1903.02428>, [arXiv:1903.02428](https://arxiv.org/abs/1903.02428)
- [8] Funke J, Tschopp FD, Grisaitis W, et al (2019) Large scale image segmentation with structured loss based deep learning for connectome reconstruction. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41(7):1669–1680. <https://doi.org/10.1109/TPAMI.2018.2835450>
- [9] Hein J, Cavalcanti N, Suter D, et al (2025) Next-generation surgical navigation: Marker-less multi-view 6DoF pose estimation of surgical instruments. *Medical Image Analysis* 103:103613. <https://doi.org/10.1016/j.media.2025.103613>
- [10] Khosla P, Teterwak P, Wang C, et al (2020) Supervised contrastive learning. In: *Advances in Neural Information Processing Systems*, pp 18661–18673
- [11] Mezger U, Jendrewski C, Bartels M (2013) Navigation in surgery. *Langenbeck’s Archives of Surgery* 398(4):501–514. <https://doi.org/10.1007/s00423-013-1059-4>
- [12] Moviglia JH, Stallkamp J (2026) Monocular near-infrared optical tracking with retroreflective fiducial markers for high-accuracy image-guided surgery. *Sensors* 26(2):357. <https://doi.org/10.3390/s26020357>
- [13] Song J, Andres B, Black MJ, et al (2019) End-to-end learning for graph decomposition. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp 10092–10101, <https://doi.org/10.1109/ICCV.2019.01019>
- [14] Strzeletz S, Hazubski S, Moctezuma JL, et al (2018) Peer-to-Peer-Navigation in der computerassistierten Chirurgie. In: Neumuth T, Melzer A, Chalopin C (eds) 17. Jahrestagung der Deutschen Gesellschaft für Computer- und Roboterassistierte Chirurgie (CURAC). Universität Leipzig, Leipzig, pp 119–124, 13–15 September 2018. ISBN 978-3-00-060786-8
- [15] Strzeletz S, Hazubski S, Moctezuma JL, et al (2020) Fast, robust, and accurate monocular peer-to-peer tracking for surgical navigation. *International Journal of Computer Assisted Radiology and Surgery* 15(3):479–489. <https://doi.org/10.1007/s11548-019-02111-z>
- [16] Tjaden H, Schwanecke U, Stein FA, et al (2015) High-speed and robust monocular tracking. In: *VISAPP 2015 — 10th International Conference on Computer Vision Theory and Applications (VISIGRAPP)*, vol 1. SciTePress, pp 462–471, <https://doi.org/10.5220/0005267104620471>
- [17] Turaga SC, Briggman KL, Helmstaedter M, et al (2009) Maximin affinity learning of image segmentation. In: *Advances in Neural Information Processing Systems*, pp 1865–1873

- [18] Wang W, Yu R, Huang Q, et al (2018) SGPN: Similarity group proposal network for 3D point cloud instance segmentation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 2569–2578, <https://doi.org/10.1109/CVPR.2018.00272>
- [19] Wang Y, Sun Y, Liu Z, et al (2019) Dynamic graph CNN for learning on point clouds. *ACM Transactions on Graphics* 38(5):146:1–146:12. <https://doi.org/10.1145/3326362>
- [20] West JB, Maurer CRJr. (2004) Designing optically tracked instruments for image-guided surgery. *IEEE Transactions on Medical Imaging* 23(5):533–545. <https://doi.org/10.1109/TMI.2004.825614>
- [21] Xu X, Yang Y, Deng C, et al (2019) Deep asymmetric metric learning via rich relationship mining. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 4076–4085, <https://doi.org/10.1109/CVPR.2019.00420>

## Supplementary material

### S1 Scope

This supplement reports the complete architecture, loss and metric definitions, experimental settings, diagnostic results, and hyperparameter optimisation (HPO) record for MoLAG. All results come from the saved benchmark and HPO artefacts.

### S2 Reproducibility details

#### S2.1 Network and training

For  $N$  detections,  $\mathcal{V} = \{1, \dots, N\}$ ,  $\mathcal{E}_{\text{mp}} = \{(j \rightarrow i) : i, j \in \mathcal{V}, i \neq j\}$ , and  $\mathcal{E}_{\text{aff}} = \{\{i, j\} : i, j \in \mathcal{V}, i < j\}$ . Thus, message passing is complete and directed, whereas affinity prediction represents each unordered pair once.

Each node begins with its normalised image coordinate  $\mathbf{h}_i^{(0)} = \tilde{\mathbf{x}}_i$ . For directed edge  $i \rightarrow j$ , the geometric attribute is

$$\mathbf{e}_{i \rightarrow j} = \left[ \|\tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_i\|_2^2, \frac{\tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_i}{\max(\|\tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_i\|_2, \varepsilon_{\text{num}})} \right]. \quad (\text{S1})$$

Each message-passing block uses the complete directed graph and applies

$$\mathbf{h}_i^{(\ell+1)} = \max_{j \neq i} \phi_{\theta_{\text{mp}}^{(\ell)}} \left( [\mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \parallel \mathbf{e}_{j \rightarrow i}] \right), \quad (\text{S2})$$

where  $\phi$  contains two linear layers with ReLU activations. The seven block widths are 128, 256, 512, 1,024, 2,048, 2,048, and 1,024. Concatenating the block outputs produces  $128 + 256 + 512 + 1,024 + 2,048 + 2,048 + 1,024 = 7,040$  features per node. For every unordered pair of detections within a scene, the affinity head receives the Hadamard product and absolute difference of the two embeddings. Its two-layer multilayer perceptron contains 128 hidden units and an intermediate ReLU. The resulting  $p_{ij} = \sigma(z_{ij})$  values weight  $\hat{\mathcal{G}}_{\text{aff}} = (\mathcal{V}, \mathcal{E}_{\text{aff}}, p)$ ; they are scene predictions rather than model parameters.

Each full-fidelity run uses 5,000,000 generated scenes for ten epochs. An additional 10,000 scenes support in-training evaluation, and the checkpoint with the lowest such evaluation loss is retained. The physical batch size is 256 scenes. Four gradient-accumulation steps yield an effective batch size of 1,024. Training uses AdamW, a learning rate of  $10^{-3}$  with cosine decay and 1 % warm-up, zero weight decay, bfloat16 arithmetic, and one NVIDIA RTX 5090 GPU. The implementation uses PyTorch 2.9.1 and PyTorch Geometric 2.7.0 [7].

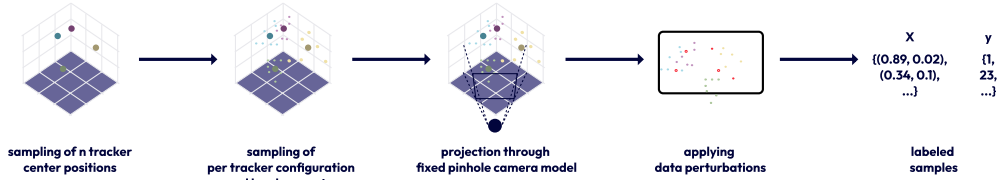
#### S2.2 Tracker, camera, pose, and localisation-error model

The evaluated rigid tracker carries seven LEDs on an equilateral triangular body with a side length of 64 mm. Three LEDs occupy the vertices. Each side has three candidate positions. Two LEDs occupy side 0 and one occupies each of sides 1 and 2. For code

$\mathbf{c} = (c_0, c_1, c_2)$ ,  $c_1$  and  $c_2$  select the occupied position on their respective sides, while  $c_0$  identifies the one omitted position on side 0. This asymmetry produces 27 distinct configurations.

The generator uses an ideal pinhole camera with a  $1440 \times 1080$  image, a  $90^\circ$  horizontal field of view, square pixels, and a centred principal point. Tracker centres are sampled with  $x, y \in [-200, 200]$  mm and  $z \in [100, 600]$  mm. A tracker’s local negative  $z$  axis first points towards the camera. It is then tilted by up to  $85^\circ$  about a random axis perpendicular to the viewing direction. Each scene contains one to ten trackers with distinct configuration codes.

Ray-triangle intersection tests model inter-tracker body occlusion. Projected LEDs behind the camera or outside the image are removed. In 80 % of scenes, LED dropout is applied independently with probability 0.4 while retaining at least three projected LEDs when possible. In 30 % of scenes, one to four spurious points are sampled within the image-space extent of the real detections. After mean centring and max-norm scaling, independent Gaussian coordinate noise with standard deviation 0.02 is applied in 80 % of scenes.



**Fig. S1** Coordinate-level sample generation. Triangular trackers are placed at sampled poses, checked for body occlusion, and projected through the camera model. LED dropout, spurious points, normalisation, and Gaussian noise then model the point-localisation interface

### S2.3 Evaluation data

Calibration uses 5,000 scenes, with 500 per tracker count. The final test suite uses 50,000 scenes, with 5,000 per tracker count. The HPO validation suite contains 20,000 scenes. Training, in-training evaluation, calibration, HPO validation, and final test data are disjoint. Calibration and final testing otherwise use the same scene modifiers and synthetic distribution as training. For full-fidelity calibration, thresholds from 0.05 to 0.95 are evaluated in steps of 0.01. The threshold with the highest strict scene grouping is retained, with ties resolved in favour of the higher threshold.

## S3 Complete loss definition

For a mini-batch of  $S$  scenes,  $V_{s,t}$  denotes the visible nodes of tracker  $t$ ,  $\mathcal{T}_s$  the trackers in scene  $s$ , and  $B_s$  its spurious nodes. The affinity head assigns logit  $z_{ij}$  to unordered pair  $\{i, j\}$ . Every condition is formed within one scene before reduction across the batch. With tracker identity  $t_i$  and negative identities for spurious points,

the conventional binary pair target is

$$y_{ij} = \mathbf{1}[t_i = t_j \wedge t_i \geq 0]. \quad (\text{S3})$$

Binary cross-entropy treats these pair labels independently. The structured objective instead uses tracker identities to express the connectivity and separation conditions of the final partition.

### S3.1 Shared reductions

For edge values  $v_a$  and non-negative handicaps  $\eta_a$ , the weighted soft maximum is

$$\mathcal{M}_\beta(v, \eta; A) = \frac{1}{\beta} \left[ \log \sum_{a \in A} \exp(\beta(v_a - \eta_a)) - \log \sum_{a \in A} \exp(-\beta \eta_a) \right]. \quad (\text{S4})$$

When at least one handicap is zero, it converges to  $\max_{a \in A} (v_a - \eta_a)$  as  $\beta \rightarrow \infty$ . Finite  $\beta$  distributes gradient among several poor edges.

If scene  $s$  contains  $n_s$  applicable conditions with losses  $q_{s,\kappa}$ , the scene reduction is

$$\mathcal{R}_{p,\rho}(q) = \frac{\sum_{s \in \mathcal{S}^*} \left( \frac{1}{n_s} \sum_{\kappa=1}^{n_s} q_{s,\kappa} \right) n_s^p}{(1 - \rho)S + \rho|\mathcal{S}^*|}, \quad \mathcal{S}^* = \{s : n_s > 0\}. \quad (\text{S5})$$

At  $p = 0$ , each scene contributes its mean condition loss; at  $p = 1$ , it contributes the sum. The parameter  $\rho \in \{0, 1\}$  determines whether scenes without an applicable condition enter the outer mean. Connectivity and separation use separate powers  $p_c$  and  $p_s$ . Bridge reduction follows  $p_c$ , whereas attachment uses  $p = 0$ .

### S3.2 Tracker connectivity

Let  $E_{s,t}^{\text{same}} = \{\{i, j\} : i, j \in V_{s,t}, i < j\}$ . A maximum spanning tree  $T_{s,t} \subseteq E_{s,t}^{\text{same}}$  is selected from the current logits, with the discrete selection detached from back-propagation. Using  $\text{sp}(x) = \log(1 + \exp x)$ ,

$$\begin{aligned} C_{s,t} &= \mathcal{M}_\beta(\psi, \eta; E_{s,t}^{\text{same}}), & \psi_{ij} &= \text{sp}(m_c - z_{ij}), \\ \eta_{ij} &= \begin{cases} 0, & \{i, j\} \in T_{s,t}, \\ \delta, & \text{otherwise,} \end{cases} & \mathcal{L}_{\text{conn}} &= \mathcal{R}_{p_c, \rho}(\{C_{s,t}\}). \end{aligned} \quad (\text{S6})$$

The discount  $\delta$  limits the contribution of non-tree edges while preserving their gradient. No connectivity condition applies to a tracker with one observed node.

### S3.3 Tracker-pair separation

For  $t \neq u$ , let  $E_{s,t,u}^{\text{diff}} = \{\{i, j\} : i \in V_{s,t}, j \in V_{s,u}\}$ . The strongest cross-tracker logit and separation loss are

$$d_{s,t,u} = \mathcal{M}_\beta(z, 0; E_{s,t,u}^{\text{diff}}), \quad \mathcal{L}_{\text{pair}} = \mathcal{R}_{p_s, \rho}(\{\text{sp}(m_s + d_{s,t,u})\}_{t < u}). \quad (\text{S7})$$

Applying the soft maximum before the margin makes its hard limit the largest cross-tracker logit.

### S3.4 Spurious attachments and bridges

For tracker  $t$  and spurious node  $b$ , define  $E_{s,t,b}^{\text{att}} = \{\{i, b\} : i \in V_{s,t}\}$ . Direct attachments are penalised by

$$\mathcal{L}_{\text{att}} = \mathcal{R}_{0, \rho}(\{\text{sp}(m_s + \mathcal{M}_\beta(z, 0; E_{s,t,b}^{\text{att}}))\}_{t,b}). \quad (\text{S8})$$

The fixed power  $p = 0$  prevents the independently sampled number of spurious points from changing scene weight.

For bridge risk, each tracker is contracted to one terminal. Set  $W(t, b) = \max_{j \in V_{s,t}} z_{bj}$  and  $W(b, b') = z_{bb'}$ . The widest path through at least one spurious node is

$$r_{s,t,u} = \max_{P: t \rightsquigarrow u, \text{int}(P) \subseteq B_s, |\text{int}(P)| \geq 1} \left( \min_{e \in P} W(e) \right), \quad (\text{S9})$$

$$\mathcal{L}_{\text{bridge}} = \mathcal{R}_{p_c, \rho}(\{\text{sp}(m_b + r_{s,t,u})\}_{t < u}).$$

A Floyd-Warshall widest-path closure computes  $r_{s,t,u}$ . If no such path exists,  $r_{s,t,u} = -\infty$  and the pair contributes zero while remaining in the scene denominator.

### S3.5 Embedding regularisation and complete objective

For real nodes  $V_s^{\text{real}} = \bigcup_t V_{s,t}$ , define normalised embeddings  $\bar{\mathbf{h}}_i = \mathbf{h}_i / \|\mathbf{h}_i\|_2$  and  $\text{sim}_{ij} = \bar{\mathbf{h}}_i^\top \bar{\mathbf{h}}_j / \tau_{\text{sc}}$ . For anchors  $\mathcal{A}_s = \{i : |P_s(i)| > 0\}$  with  $P_s(i) = \{j \in V_s^{\text{real}} \setminus \{i\} : t_j = t_i\}$ ,

$$\mathcal{L}_{\text{supcon}} = \frac{1}{|\mathcal{S}_{\text{sc}}|} \sum_{s \in \mathcal{S}_{\text{sc}}} \frac{1}{|\mathcal{A}_s|} \sum_{i \in \mathcal{A}_s} \left[ \log \sum_{j \in V_s^{\text{real}} \setminus \{i\}} e^{\text{sim}_{ij}} - \frac{1}{|P_s(i)|} \sum_{j \in P_s(i)} \text{sim}_{ij} \right], \quad (\text{S10})$$

where  $\mathcal{S}_{\text{sc}} = \{s : |\mathcal{A}_s| > 0\}$ . Spurious nodes and cross-scene pairs are excluded. The complete family is

$$\mathcal{L} = w_c \mathcal{L}_{\text{conn}} + w_s (\mathcal{L}_{\text{pair}} + \varepsilon_{\text{spur}} \mathcal{L}_{\text{att}}) + w_b \mathcal{L}_{\text{bridge}} + w_{sc} \mathcal{L}_{\text{supcon}}. \quad (\text{S11})$$

The combined full-fidelity setting uses  $w_c = 1$ ,  $w_s = 0.46$ ,  $w_b = 0.25$ ,  $m_c = m_s = 1$ ,  $m_b = 0$ ,  $\beta = 1$ ,  $\delta = 3$ ,  $\varepsilon_{\text{spur}} = 0.01$ ,  $p_c = p_s = 0.5$ ,  $\rho = 1$ , and  $w_{sc} = 0$ .

## S4 Complete metric definitions

Let  $\mathcal{S}_{\text{test}}$  contain  $N$  scenes. For scene  $s$ ,  $\mathcal{V}_s$  is the node set,  $V_{s,t}$  the nodes of tracker  $t$ ,  $\mathcal{T}_s$  the tracker set,  $B_s$  the spurious nodes, and  $\hat{c}_s(i)$  the predicted component of node  $i$ . The retained affinity edges are  $\mathcal{E}_{\tau,s} = \{\{i, j\} : p_{ij} \geq \tau\}$ .

### S4.1 Strict grouping and recovery

The unsplit, unmerged, and strict-recovery indicators for tracker  $t$  are

$$\begin{aligned} u_{s,t} &= \mathbf{1}[|\{\hat{c}_s(i) : i \in V_{s,t}\}| = 1], \\ m_{s,t} &= \mathbf{1}[\nexists t' \neq t, i \in V_{s,t}, j \in V_{s,t'} : \hat{c}_s(i) = \hat{c}_s(j)], \\ r_{s,t} &= u_{s,t} m_{s,t}. \end{aligned} \quad (\text{S12})$$

Thus, strict recovery requires all visible LEDs of one tracker to share a component that contains no real LED from another tracker. Spurious detections may attach to one tracker without failure; they cause failure when they connect trackers. Strict scene grouping and all-tracker strict recovery are

$$g_s = \prod_{t \in \mathcal{T}_s} r_{s,t}, \quad A_{\text{group}} = \frac{1}{N} \sum_s g_s, \quad R_{\text{all}}^{\text{strict}} = \frac{\sum_s \sum_{t \in \mathcal{T}_s} r_{s,t}}{\sum_s |\mathcal{T}_s|}. \quad (\text{S13})$$

Removing every edge incident to a spurious node defines component assignment  $\hat{c}_s^{\text{real}}$  and corresponding indicator  $r_{s,t}^{\text{real}}$ . Real-only grouping is  $A_{\text{group}}^{\text{real}} = N^{-1} \sum_s \prod_{t \in \mathcal{T}_s} r_{s,t}^{\text{real}}$ .

### S4.2 Complete-tracker measures

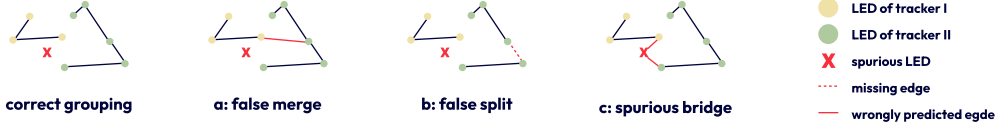
Let  $\mathcal{T}_s^{(\tau)} = \{t : |V_{s,t}| = 7\}$ . The complete-tracker share and strict recovery rate are

$$Q_{\text{complete}} = \frac{\sum_s |\mathcal{T}_s^{(\tau)}|}{\sum_s |\mathcal{T}_s|}, \quad R_{\text{complete}}^{\text{strict}} = \frac{\sum_s \sum_{t \in \mathcal{T}_s^{(\tau)}} r_{s,t}}{\sum_s |\mathcal{T}_s^{(\tau)}|}. \quad (\text{S14})$$

For an unsplit tracker, let  $\hat{c}_{s,t}$  be its single component. Extractability counts each component containing at least one unsplit complete tracker once:

$$E_{\text{complete}} = \frac{\sum_s \left| \left\{ \hat{c}_{s,t} : t \in \mathcal{T}_s^{(\tau)}, u_{s,t} = 1 \right\} \right|}{\sum_s |\mathcal{T}_s^{(\tau)}|}. \quad (\text{S15})$$

A split tracker contributes zero. If two complete trackers share a component, that component contributes once.



**Fig. S2** Principal grouping outcomes. Node colour denotes the true tracker, and the cross denotes a spurious detection. Dark edges are correct affinities; solid red edges create merges and the dashed red edge creates a split. A false merge can fail strict grouping while retaining an extractable complete tracker

### S4.3 Edge and failure diagnostics

For pair label  $y_{ij} = \mathbf{1}[i, j \notin B_s \wedge t_i = t_j]$  and prediction  $\hat{y}_{ij} = \mathbf{1}[p_{ij} \geq \tau]$ , aggregate true positives, false positives, and false negatives over edge set  $A$ :

$$P_A = \frac{\text{TP}_A}{\text{TP}_A + \text{FP}_A}, \quad R_A = \frac{\text{TP}_A}{\text{TP}_A + \text{FN}_A}, \quad F_{1,A} = \frac{2P_A R_A}{P_A + R_A}. \quad (\text{S16})$$

All-pair metrics use every unordered within-scene pair. Real-real metrics include only pairs of true LEDs.

Define scene split and merge indicators  $S_s = 1 - \prod_t u_{s,t}$  and  $M_s = 1 - \prod_t m_{s,t}$ . Let  $M_s^{\text{real}}$  use components after removing spurious-incident edges, and let  $q_s^{\text{bridge}} = M_s(1 - M_s^{\text{real}})$ . The mutually exclusive outcomes are

$$\begin{aligned} I_s^{\text{correct}} &= (1 - S_s)(1 - M_s), & I_s^{\text{split}} &= S_s(1 - M_s), \\ I_s^{\text{mixed}} &= S_s M_s, & I_s^{\text{bridge}} &= (1 - S_s) q_s^{\text{bridge}}, \\ I_s^{\text{merge}} &= (1 - S_s) M_s (1 - q_s^{\text{bridge}}). \end{aligned} \quad (\text{S17})$$

Each rate is  $N^{-1} \sum_s I_s$ . The broader spurious-bridge rate  $N^{-1} \sum_s q_s^{\text{bridge}}$  also includes scenes with splits.

The direct spurious-attachment rate is

$$A_{\text{spur}} = \frac{\sum_s \sum_{b \in B_s} \mathbf{1}[\exists i \in \mathcal{V}_s \setminus B_s : \{b, i\} \in \mathcal{E}_{\tau,s}]}{\sum_s |B_s|}. \quad (\text{S18})$$

It is descriptive rather than an error by itself.

### S4.4 Statistical procedures

Strict scene grouping uses 95 % Wilson score intervals. Complete-tracker strict recovery and extractability use scene-clustered ratio intervals with a delta-method variance estimate. Full-fidelity checkpoint comparisons use paired McNemar tests on strict scene grouping and the scene event that every complete tracker is extractable; signed  $z$  statistics use a two-sided 5 % level. Each full-fidelity configuration was trained once, so these tests quantify finite-test-suite uncertainty rather than training variation.

Each Phase-B HPO arm comprises three runs and is summarised by its mean and sample standard deviation. The primary selected-candidate comparison also retains an earlier incumbent run and pools within-arm variance across four arms to form a two-sided  $t$  interval. Secondary comparisons use the three Phase-B runs per arm. All HPO inference is limited to 6 % training fidelity.

## S5 Detailed full-fidelity results

### S5.1 Loss configurations and paired comparisons

Table S1 reports the absolute result for each full-fidelity checkpoint. Every row uses one training run and the same frozen test suite. The combined scaled-conjunction checkpoint achieved the highest strict grouping, complete-tracker extractability, real-real edge  $F_1$ , and the lowest spurious attachment rate among the six settings.

**Table S1** Full-fidelity results. SG: strict scene grouping; Ext.: complete-tracker extractability; CSR: complete-tracker strict recovery; ASR: all-tracker strict recovery;  $F_1$ : real-real edge  $F_1$ ; SA: spurious-attachment rate. All entries after the threshold are percentages

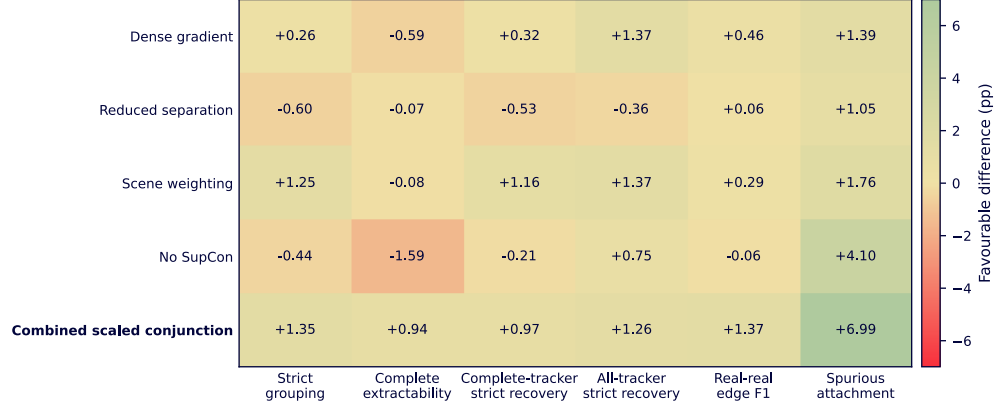
| Setting                     | Threshold | SG     | Ext.   | CSR    | ASR    | $F_1$  | SA     |
|-----------------------------|-----------|--------|--------|--------|--------|--------|--------|
| Conjunction reference       | 0.71      | 55.078 | 88.699 | 79.393 | 74.946 | 90.967 | 75.012 |
| Dense gradient              | 0.77      | 55.336 | 88.106 | 79.709 | 76.317 | 91.425 | 73.623 |
| Reduced separation          | 0.88      | 54.482 | 88.631 | 78.864 | 74.584 | 91.025 | 73.957 |
| Scene weighting             | 0.76      | 56.330 | 88.623 | 80.557 | 76.312 | 91.259 | 73.250 |
| No SupCon                   | 0.78      | 54.638 | 87.105 | 79.184 | 75.697 | 90.908 | 70.913 |
| Combined scaled conjunction | 0.84      | 56.426 | 89.642 | 80.360 | 76.204 | 92.339 | 68.020 |

Table S2 reports rate differences and paired tests against the conjunction reference. The extractability difference is the tracker-level rate change. Its paired test instead uses the scene-level event that every complete tracker in a scene is extractable. The tests account for the shared frozen scenes. They do not include variation between training runs and are therefore checkpoint-level comparisons.

**Table S2** Full-fidelity checkpoint comparisons with the conjunction reference. Differences are percentage-point changes in strict grouping and tracker-level extractability. Signed McNemar  $z$  statistics use the corresponding binary scene outcomes on the common frozen test suite

| Setting                     | $\Delta$ SG | $z_{SG}$ | $\Delta$ Ext. | $z_{Ext.}$ |
|-----------------------------|-------------|----------|---------------|------------|
| Dense gradient              | +0.258      | +2.25    | -0.593        | -9.04      |
| Reduced separation          | -0.596      | -6.13    | -0.068        | -1.85      |
| Scene weighting             | +1.252      | +12.13   | -0.075        | -1.46      |
| No SupCon                   | -0.440      | -4.25    | -1.594        | -14.32     |
| Combined scaled conjunction | +1.348      | +12.58   | +0.944        | +5.03      |

Figure S3 extends the main two-metric comparison. It shows how each setting changes the two strict-recovery rates and edge behaviour relative to the reference. The combined setting improves all displayed measures, with the largest change in spurious attachment.



**Fig. S3** Full-fidelity differences from the conjunction reference. Positive values are favourable. The sign of spurious attachment is reversed because a lower rate is better. Each row is one trained checkpoint evaluated on the common frozen test suite

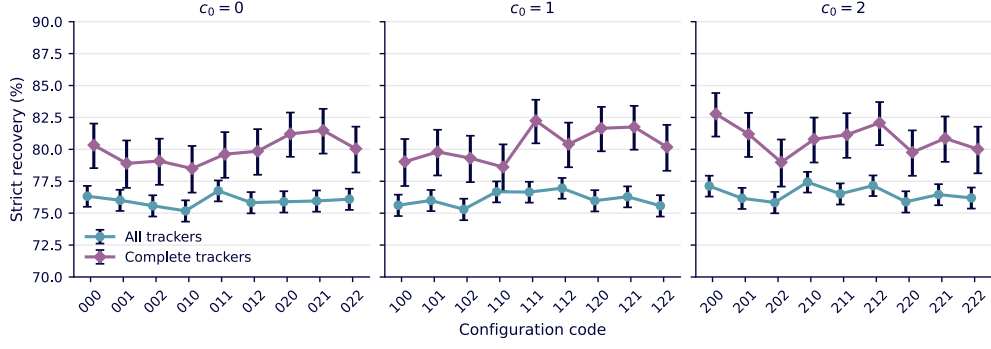
## S5.2 Diagnostic results

Table S3 reports the mutually exclusive outcome counts for the combined checkpoint. Merges and mixed errors account for most incorrect scenes. Only 66 scenes fail solely through a path of spurious detections.

**Table S3** Mutually exclusive outcomes for the 50,000-scene frozen test suite

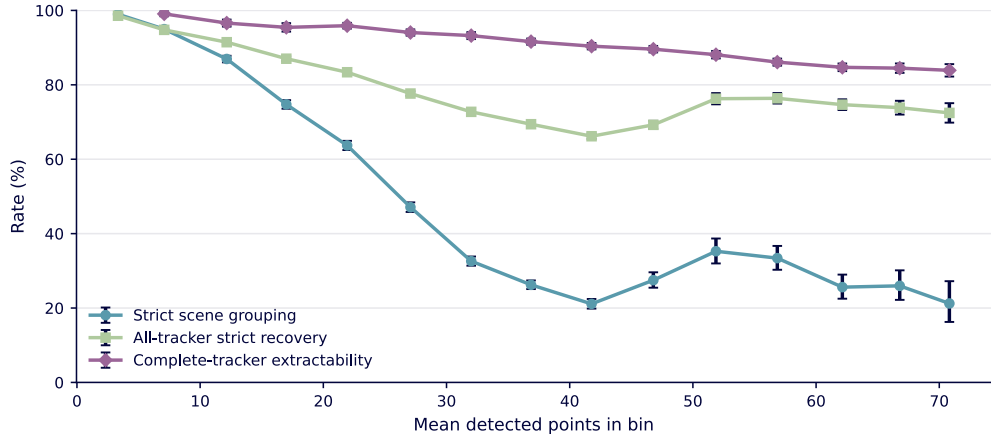
| Outcome               | Scenes | Rate (%) |
|-----------------------|--------|----------|
| Correct grouping      | 28,213 | 56.426   |
| False split           | 1,276  | 2.552    |
| False merge           | 10,511 | 21.022   |
| Mixed split and merge | 9,934  | 19.868   |
| Spurious bridge only  | 66     | 0.132    |

The 27 tracker configuration codes occur at comparable frequency. All-tracker and complete-tracker strict recovery are also consistent across the codes (Figure S4). Complete-tracker intervals are wider because complete trackers form the smaller subset.



**Fig. S4** All-tracker and complete-tracker strict recovery across the 27 configuration codes. A three-digit label identifies  $(c_0, c_1, c_2)$ , and panels group configurations by  $c_0$ . Blue circles include all tracker instances. Purple diamonds include only complete seven-LED trackers. Whiskers are 95% Wilson intervals

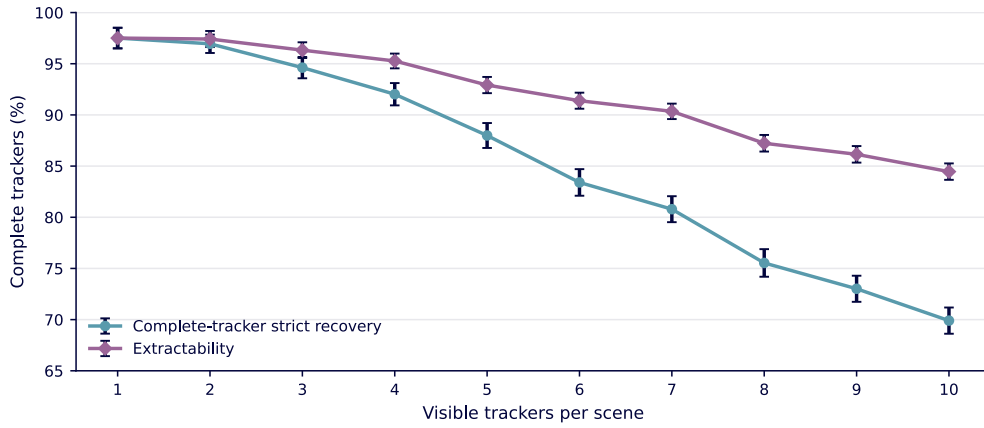
Performance by total detected-point count provides a complementary view of scene complexity (Figure S5). Strict scene grouping falls as inputs grow. All-tracker strict recovery decreases more gradually, while complete-tracker extractability remains high across the observed range. Point count also changes with tracker count, visibility, and clutter, so this stratification is descriptive.



**Fig. S5** Performance by total detected-point count for the combined configuration. Scenes are grouped into five-point bins, and each horizontal coordinate is the observed mean within its bin. Strict scene grouping uses Wilson intervals. All-tracker strict recovery and complete-tracker extractability use scene-clustered ratio intervals

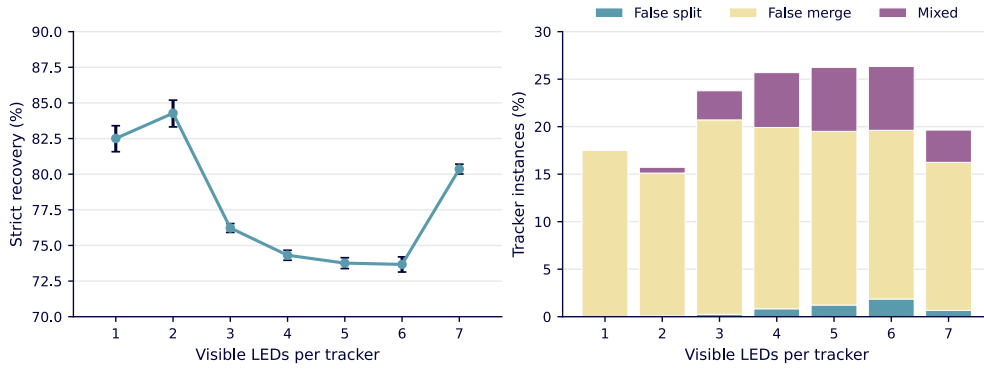
Complete-tracker strict recovery and extractability diverge with crowding because extractability accepts an intact tracker inside a merged component (Figure S6).

Strict recovery varies non-monotonically with visible LEDs and generally decreases with crowding (Figures S7 and S8). Complete seven-LED trackers and sparse one-



**Fig. S6** Complete-tracker outcomes by visible tracker count. Strict recovery requires an unsplit and unmerged tracker; extractability also accepts an intact tracker in a merged component. Whiskers are 95 % scene-clustered ratio intervals

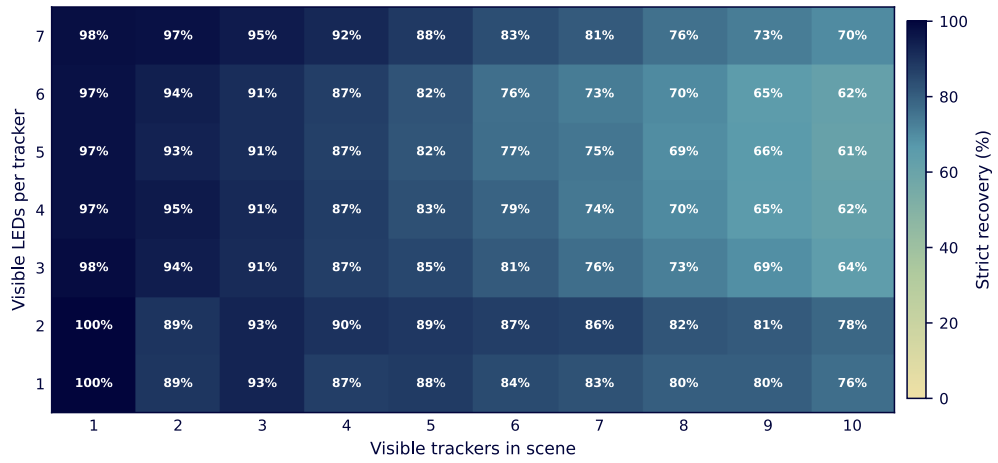
or two-LED cases have higher recovery than the intermediate visibility levels; false merges dominate throughout.



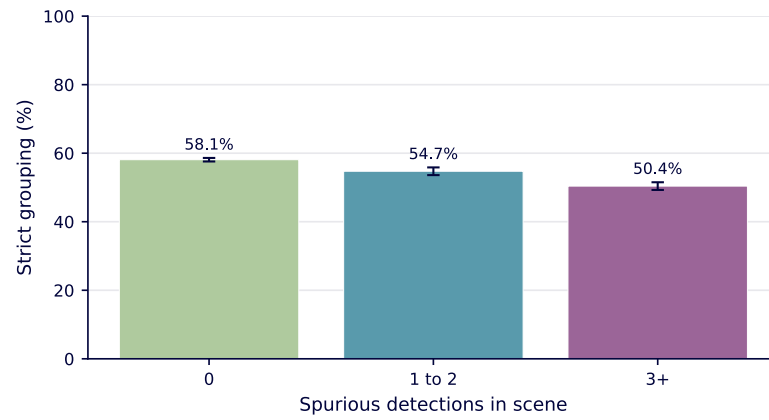
**Fig. S7** Tracker-level outcomes by visible LEDs. The panels show strict recovery and mutually exclusive false-split, false-merge, and mixed-error rates

Strict grouping decreases with spurious detections, although spurious-bridge-only failures remain rare (Figure S9).

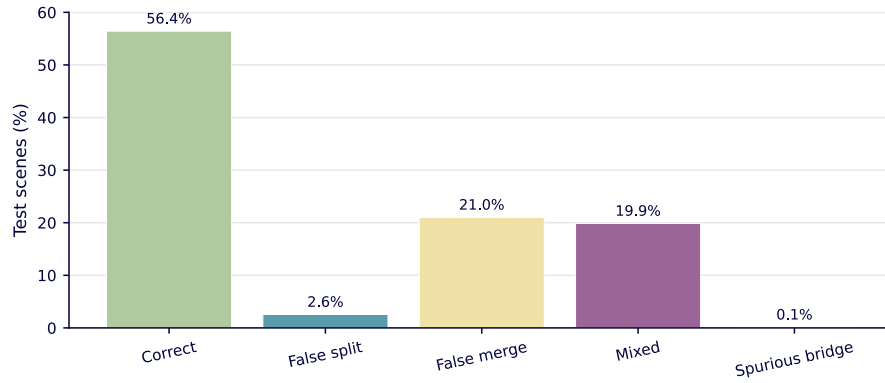
Merges and mixed errors dominate incorrect partitions and increase with tracker count (Figures S10 and S11).



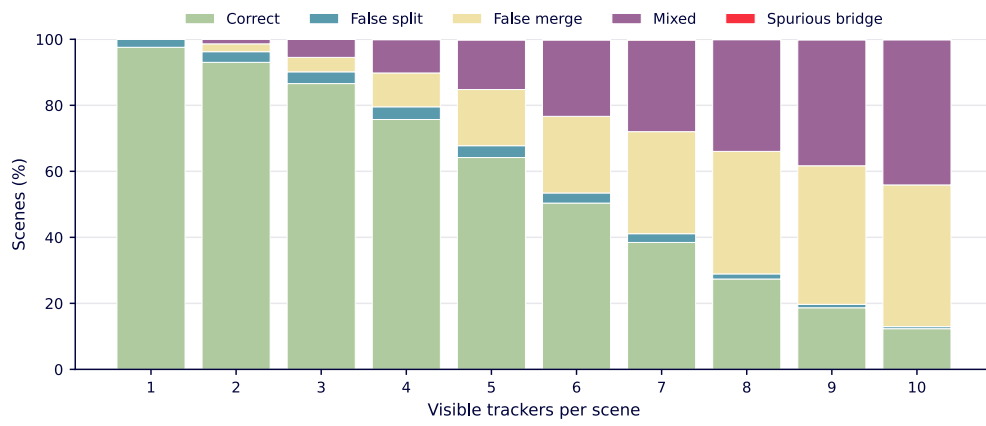
**Fig. S8** Strict recovery by visible tracker count and visible LEDs per tracker. Cell labels give percentages



**Fig. S9** Strict scene grouping by spurious-detection count. Labels give rates and whiskers show 95% Wilson intervals



**Fig. S10** Mutually exclusive outcomes for the 50,000 frozen scenes. Bar labels give percentages



**Fig. S11** Mutually exclusive scene outcomes by exact visible tracker count. Each bar represents 5,000 scenes and sums to 100 %

## S6 Additional scope considerations

The synthetic distribution is a controlled stress model rather than a measured model of an operating room. In 80 % of scenes, dropout uses a pre-constraint per-LED probability of 0.4 and retains at least three projected LEDs when possible. Projection can still leave one or two LEDs. Physical occlusion is spatially structured by the camera, trackers, instruments, and anatomy, whereas the post-projection dropout is random. Without measured localisation data, the relative difficulty of these effects in deployment is unknown.

Spurious points occur in 30 % of scenes and are sampled uniformly inside the extent of real detections. This tests nearby clutter but not reflections or the appearance distribution of false detections. Independent Gaussian noise does not model camera distortion, detector thresholds, merged detections, motion, or calibration error. The ideal pinhole camera, stylised poses, distinct within-scene codes, triangular geometry, and removal of global position and scale further delimit the generalisation claim.

Each sample represents one independent frame. Temporal association, camera fusion, synchronisation, communication, and motion-dependent localisation are not evaluated. Runtime and pose performance also remain unmeasured. Extractability assumes that the downstream method can identify a complete tracker among extra detections, and a corresponding geometric procedure for the triangular tracker has not yet been tested. Recorded observations from several peer cameras are also needed to determine whether complementary views improve tracker availability when one view is incomplete.

## S7 Exploratory HPO study

The HPO had three stages. Phase 0 checked whether short runs reproduced the ordering of known full-fidelity configurations. Phase A searched the loss space at 6 % training fidelity. Phase B repeated selected settings and measured variation between training runs.

### S7.1 Design and proxy check

**Table S4** HPO stages and their roles

| Stage   | Completed work             | Fidelity | Role                           |
|---------|----------------------------|----------|--------------------------------|
| Phase 0 | Seven known settings       | 3 %      | Initial proxy check            |
| Phase A | 77 search settings         | 6 %      | Candidate search               |
| Phase B | Four arms, three runs each | 6 %      | Replication and noise estimate |

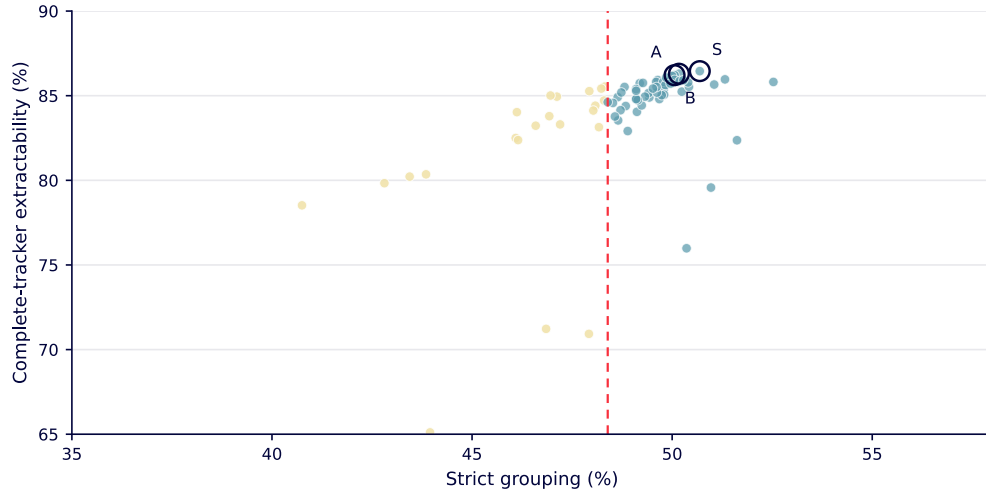
The Phase-0 check was uninformative. Across the six conjunction-family settings, the Spearman correlation with the full-fidelity ordering was  $-0.486$  at 3 % fidelity. At 6 %, it was  $-0.600$  with the single incumbent run and  $0.086$  after replacing that

observation with the later repeat mean. These unstable estimates from six settings do not validate proxy rankings. The HPO is therefore interpreted through its direct Phase-B replication at 6 % fidelity.

Each Phase-A setting trained on 1,000,000 generated scenes for three epochs. Even-indexed validation scenes formed split A for threshold selection. Odd-indexed scenes formed split B for scoring. The objective was complete-tracker extractability on split B, with strict grouping of at least 0.4839. Fifty-four of the 77 completed settings met this final floor.

## S7.2 Search patterns and selected settings

Figure S12 shows the objective and constraint for every completed search setting. The three ringed settings were retained for replication because they represented distinct parts of the high-extractability region.



**Fig. S12** Complete-tracker extractability and strict grouping for 77 completed search settings at 6 % fidelity. Blue points meet the final strict-grouping floor of 48.39 %, while yellow points fall below it. The dashed line marks the floor. Rings identify candidate A, candidate B, and the selected candidate

All 20 highest-scoring feasible settings used finite soft maximum aggregation. Table S5 summarises the other shared patterns. The stronger settings apply more scene scaling to separation and use a lower separation weight.

Table S6 reports the exact retained settings. Candidate labels are used in the figures and main text. Trial numbers link them to the complete Phase-A record. Each setting uses eligible-scene averaging and finite soft maximum aggregation. The non-tree discount is  $d = \exp(-\beta\delta)$ .

**Table S5** Median parameters among the 20 highest- and 20 lowest-scoring feasible Phase-A settings

| Parameter                      | Highest 20 | Lowest 20 |
|--------------------------------|------------|-----------|
| Connectivity scene power $p_c$ | 0.626      | 0.379     |
| Separation scene power $p_s$   | 0.887      | 0.503     |
| Separation weight $w_s$        | 0.285      | 0.569     |

**Table S6** Exact parameters of the three settings retained for replication. A zero SupCon weight disables that term

| Candidate           | $m_c$   | $m_s$   | $p_c$   | $p_s$   | $w_s$   | $\beta$  | $d$     | $w_{sc}$ |
|---------------------|---------|---------|---------|---------|---------|----------|---------|----------|
| A (trial 36)        | 1.93439 | 3.60479 | 0.66497 | 0.87918 | 0.28686 | 56.26569 | 0.03105 | 0.054485 |
| B (trial 41)        | 0.50000 | 0.50000 | 0.73415 | 1.00000 | 0.45690 | 5.91172  | 0.00411 | 0        |
| Selected (trial 71) | 0.50000 | 4.00000 | 0.57319 | 0.87674 | 0.25780 | 3.11608  | 0.22098 | 0.009626 |

### S7.3 Replication and measured training variation

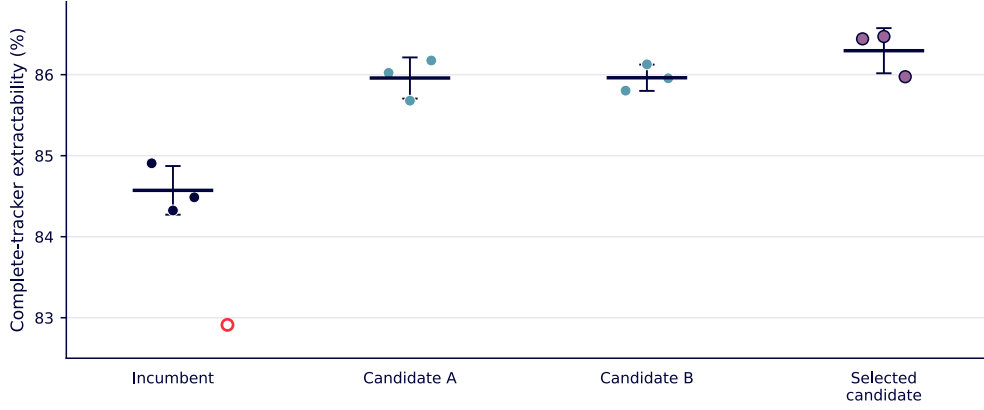
The three selected candidates replicated within a narrow range and all exceeded the three-run incumbent mean (Table S7). The similarity between the candidates supports a high-performing region rather than a unique optimum.

**Table S7** Complete-tracker extractability on split B for Phase-B runs at 6 % fidelity. Values are percentages. SD is the within-arm standard deviation in percentage points

| Arm                    | Run 1   | Run 2   | Run 3   | Mean    | SD     |
|------------------------|---------|---------|---------|---------|--------|
| Proxy incumbent        | 84.9060 | 84.3240 | 84.4862 | 84.5721 | 0.3004 |
| Candidate A (trial 36) | 86.0223 | 85.6788 | 86.1750 | 85.9587 | 0.2541 |
| Candidate B (trial 41) | 85.8029 | 86.1273 | 85.9555 | 85.9619 | 0.1623 |
| Selected (trial 71)    | 86.4421 | 86.4708 | 85.9746 | 86.2958 | 0.2785 |

The primary comparison retains an earlier same-fidelity incumbent observation of 82.9119 %. The four-run incumbent mean is 84.1570 %. Pooling within-arm variation across the four replicated arms after retaining this observation yields  $\sigma = 0.5359$  percentage points with 9 degrees of freedom. The selected candidate exceeds the incumbent by 2.1388 percentage points. The two-sided test yields  $t = 5.226$ ,  $p = 5.5 \times 10^{-4}$ , and a 95 % interval of [1.213, 3.065] percentage points. The interval excludes zero after accounting for the measured between-run variation. This noise estimate and inference apply only to the 6 % fidelity design.

Table S8 shows the broader operating-point change. The selected candidate improves both complete-tracker measures. Strict scene grouping and all-tracker strict



**Fig. S13** Replicated complete-tracker extractability after 6% fidelity training. Points show three runs per arm; bars and whiskers show the mean and sample standard deviation. The open point is the earlier incumbent observation retained in the primary comparison

recovery remain similar. Higher real-real recall is accompanied by lower precision, lower edge  $F_1$ , and more spurious attachment.

**Table S8** Three-run sensitivity comparison of the selected candidate with the proxy-fidelity incumbent. Differences are percentage points. The reported  $p$  values use the pooled three-run analysis; the primary extractability inference is reported in the paragraph above

| Metric                           | Incumbent (%) | Selected (%) | Difference | Inference            |
|----------------------------------|---------------|--------------|------------|----------------------|
| Complete-tracker extractability  | 84.57         | 86.30        | +1.72      | Primary result above |
| Complete-tracker strict recovery | 70.76         | 72.66        | +1.91      | $p = 0.005$          |
| Strict scene grouping            | 49.91         | 50.40        | +0.49      | $p = 0.19$           |
| All-tracker strict recovery      | 71.81         | 71.91        | +0.11      | $p = 0.77$           |
| Real-real edge $F_1$             | 88.76         | 87.96        | -0.80      | $p = 10^{-4}$        |
| Spurious attachment              | 70.40         | 78.30        | +7.90      | $p < 10^{-7}$        |

## S7.4 Complete Phase-A record

Table S9 lists every completed search setting. The separation and SupCon weights are shown as compact search coordinates; the search also varied the parameters in Table S6.

**Table S9:** Complete Phase-A search record. Grouping is strict scene grouping on split B

| Trial | Objective | Grouping (%) | Threshold | Feasible | $w_s$  | $w_{sc}$ |
|-------|-----------|--------------|-----------|----------|--------|----------|
| 0     | 0.840473  | 49.12        | 0.675     | Yes      | 2.0000 | 0.03000  |
| 1     | 0.835416  | 48.65        | 0.750     | Yes      | 2.0000 | 0.03000  |
| 2     | 0.831409  | 48.17        | 0.875     | No       | 0.9000 | 0.03000  |

| Trial | Objective | Grouping (%) | Threshold | Feasible | $w_s$  | $w_{sc}$ |
|-------|-----------|--------------|-----------|----------|--------|----------|
| 3     | 0.852400  | 50.24        | 0.700     | Yes      | 1.0000 | 0.03000  |
| 4     | 0.844290  | 49.24        | 0.675     | Yes      | 2.0000 | 0.00000  |
| 5     | 0.829119  | 48.89        | 0.850     | Yes      | 0.4600 | 0.00000  |
| 6     | 0.848011  | 49.10        | 0.950     | Yes      | 0.3641 | 0.03888  |
| 7     | 0.843908  | 48.84        | 0.950     | Yes      | 0.3077 | 0.16542  |
| 8     | 0.858315  | 50.15        | 0.850     | Yes      | 0.3409 | 0.11884  |
| 9     | 0.853831  | 49.79        | 0.625     | Yes      | 0.3579 | 0.00000  |
| 10    | 0.759851  | 50.36        | 0.400     | Yes      | 0.6480 | 0.00678  |
| 11    | 0.849442  | 47.12        | 0.950     | No       | 0.2112 | 0.03156  |
| 12    | 0.712241  | 46.85        | 0.400     | No       | 3.0000 | 0.00404  |
| 13    | 0.855357  | 48.31        | 0.950     | No       | 0.1415 | 0.00000  |
| 14    | 0.849156  | 48.64        | 0.575     | Yes      | 0.2221 | 0.00000  |
| 15    | 0.803549  | 43.85        | 0.950     | No       | 0.5784 | 0.16152  |
| 16    | 0.850110  | 46.96        | 0.950     | No       | 0.1053 | 0.00935  |
| 17    | 0.848965  | 49.43        | 0.400     | Yes      | 0.8095 | 0.21193  |
| 18    | 0.709283  | 47.92        | 0.400     | No       | 1.2731 | 0.00963  |
| 19    | 0.841427  | 48.71        | 0.775     | Yes      | 2.0810 | 0.00000  |
| 20    | 0.795726  | 50.97        | 0.400     | Yes      | 0.4899 | 0.00000  |
| 21    | 0.859651  | 51.32        | 0.850     | Yes      | 0.3122 | 0.24968  |
| 22    | 0.845721  | 48.52        | 0.900     | Yes      | 0.2318 | 0.00000  |
| 23    | 0.637535  | 42.20        | 0.400     | No       | 3.0000 | 0.05899  |
| 24    | 0.847152  | 48.30        | 0.900     | No       | 0.1502 | 0.07877  |
| 25    | 0.857361  | 49.19        | 0.525     | Yes      | 0.1393 | 0.00000  |
| 26    | 0.798302  | 42.81        | 0.950     | No       | 0.2327 | 0.00000  |
| 27    | 0.859269  | 49.64        | 0.475     | Yes      | 0.2513 | 0.00000  |
| 28    | 0.651083  | 43.95        | 0.400     | No       | 0.7549 | 0.01710  |
| 29    | 0.846198  | 48.39        | 0.925     | Yes      | 0.0723 | 0.00000  |
| 30    | 0.858124  | 49.60        | 0.925     | Yes      | 0.4700 | 0.17110  |
| 31    | 0.858220  | 49.78        | 0.525     | Yes      | 0.2106 | 0.00000  |
| 32    | 0.851541  | 49.41        | 0.750     | Yes      | 0.2203 | 0.00000  |
| 36    | 0.862990  | 50.17        | 0.825     | Yes      | 0.2869 | 0.05449  |
| 37    | 0.840282  | 46.12        | 0.950     | No       | 0.1364 | 0.00000  |
| 38    | 0.855166  | 48.81        | 0.750     | Yes      | 0.1855 | 0.00000  |
| 39    | 0.858983  | 49.94        | 0.675     | Yes      | 0.2514 | 0.00000  |
| 40    | 0.832268  | 46.59        | 0.950     | No       | 0.0704 | 0.00930  |
| 41    | 0.862131  | 50.06        | 0.525     | Yes      | 0.4569 | 0.00000  |
| 42    | 0.853926  | 49.10        | 0.750     | Yes      | 0.2664 | 0.00442  |
| 43    | 0.823681  | 51.62        | 0.400     | Yes      | 0.6477 | 0.00000  |
| 44    | 0.861654  | 50.00        | 0.575     | Yes      | 0.3295 | 0.00000  |
| 45    | 0.851923  | 48.73        | 0.775     | Yes      | 0.1166 | 0.00000  |
| 46    | 0.856311  | 49.83        | 0.475     | Yes      | 0.6491 | 0.00000  |
| 47    | 0.857743  | 50.02        | 0.600     | Yes      | 0.6134 | 0.16542  |
| 48    | 0.850587  | 49.80        | 0.825     | Yes      | 0.6992 | 0.13797  |
| 49    | 0.848011  | 49.68        | 0.475     | Yes      | 2.6459 | 0.23740  |
| 50    | 0.837706  | 48.57        | 0.600     | Yes      | 3.0000 | 0.03023  |
| 51    | 0.860700  | 49.85        | 0.575     | Yes      | 0.2402 | 0.00000  |
| 52    | 0.825017  | 46.09        | 0.575     | No       | 0.0500 | 0.00000  |
| 53    | 0.849442  | 49.32        | 0.750     | Yes      | 0.2052 | 0.00000  |
| 54    | 0.850301  | 49.74        | 0.825     | Yes      | 0.1960 | 0.00000  |
| 55    | 0.844099  | 48.08        | 0.950     | No       | 0.0926 | 0.03131  |
| 56    | 0.833031  | 47.20        | 0.925     | No       | 0.5002 | 0.00000  |
| 57    | 0.847343  | 49.14        | 0.900     | Yes      | 0.1711 | 0.08862  |
| 58    | 0.851827  | 49.62        | 0.850     | Yes      | 0.2215 | 0.00000  |
| 59    | 0.841237  | 48.03        | 0.700     | No       | 0.2220 | 0.00000  |
| 60    | 0.848011  | 49.10        | 0.450     | Yes      | 0.7989 | 0.13176  |
| 61    | 0.855166  | 49.61        | 0.675     | Yes      | 0.1571 | 0.00000  |
| 62    | 0.855262  | 50.42        | 0.650     | Yes      | 1.3017 | 0.01452  |
| 63    | 0.823776  | 46.15        | 0.775     | No       | 3.0000 | 0.00000  |
| 64    | 0.859365  | 50.06        | 0.850     | Yes      | 0.2838 | 0.00000  |
| 65    | 0.837897  | 46.93        | 0.875     | No       | 0.0500 | 0.00000  |
| 66    | 0.857838  | 50.40        | 0.950     | Yes      | 0.2667 | 0.00000  |
| 67    | 0.852877  | 49.10        | 0.675     | Yes      | 0.3250 | 0.00000  |

| Trial | Objective | Grouping (%) | Threshold | Feasible | $w_s$  | $w_{sc}$ |
|-------|-----------|--------------|-----------|----------|--------|----------|
| 68    | 0.854212  | 49.52        | 0.550     | Yes      | 0.1645 | 0.00000  |
| 69    | 0.856598  | 51.05        | 0.700     | Yes      | 0.4205 | 0.00550  |
| 70    | 0.857552  | 49.27        | 0.950     | Yes      | 0.3072 | 0.01219  |
| 71    | 0.864517  | 50.69        | 0.525     | Yes      | 0.2578 | 0.00963  |
| 72    | 0.802214  | 43.44        | 0.950     | No       | 0.0500 | 0.00000  |
| 73    | 0.852686  | 47.93        | 0.450     | No       | 0.1541 | 0.00000  |
| 74    | 0.857170  | 49.96        | 0.600     | Yes      | 0.1710 | 0.00000  |
| 75    | 0.858124  | 52.53        | 0.400     | Yes      | 0.3275 | 0.00000  |
| 76    | 0.858697  | 50.19        | 0.725     | Yes      | 0.1422 | 0.00000  |
| 77    | 0.785230  | 40.75        | 0.950     | No       | 0.0500 | 0.02503  |
| 78    | 0.859174  | 50.28        | 0.775     | Yes      | 0.3046 | 0.00000  |
| 79    | 0.854212  | 48.23        | 0.750     | No       | 0.0936 | 0.00000  |

### S7.5 Interpretive scope

Phase-A selection and Phase-B replication both use split B. Replication measures training variation but does not remove all selection optimism. The parity split is not exactly stratified. The feasibility floor was updated and then applied to the complete search record; one high-objective setting lies 0.0008 below it. Threshold search was bounded to 0.40 to 0.95. A fresh-split protocol was prepared but did not produce a completed result in this study.

The margin parameters have a near-gauge freedom. Their sampled values therefore do not establish their importance. Conditional parameters were sampled randomly when enabled, so they do not locate an optimum. Most importantly, no HPO-selected setting was trained with the full-fidelity recipe. The replicated result establishes a high-extractability operating region at 6 % fidelity.